Natural Language Processing

(TEE7101 / 0701260505)

Theory | Credits: 3 | No. of Hours: 45 | Lectures/Week: 3

Internal: 30M | External (ESE): 45M

SIT Pune | AIML | B.Tech 2024-28 | Semester V

Evaluation Plan

Assessment is linked with course outcomes

CA1 Quiz

CO1

Unit 1: Introduction to Language

10 marks scaled to 4

Tentative: 03/08/2026

CA2 Unit Test

CO2 and CO3

Unit 2: Feature Extraction

Unit 3: Word Embeddings

20 marks scaled to 15

Tentative: 07/09/2026 to 12/09/2026

CA3 Case Studies

CO4 and CO5

Advanced NLP models

Machine translation

12 marks

Tentative: 22/10/2026

Tip: build a small NLP example portfolio while learning. It makes CA3 easier.

Why NLP Now?

Language is the main interface to modern AI

You already use NLP when you...

- search with natural questions
- translate messages instantly
- dictate to your phone
- get auto-captions in videos
- ChatGPT, Gemini, Copilot, or chatbots

Industry uses NLP for...

- customer-support automation
- resume and document screening
- medical and legal document analysis
- sentiment and social media analytics
- knowledge assistants over company data

If a machine answers in fluent English, does it really understand language?

What Is NLP?

Natural Language Processing helps machines work with human language

NLP is the AI area that allows computers to read, write, listen, speak, translate, search, summarize, and answer using natural language.

Examples

- Google Search understanding a question
- YouTube auto captions
- WhatsApp predictive text
- Google Translate
- ChatGPT / Gemini / Copilot

Key challenge

- Human language is ambiguous
- Meaning depends on context
- Same word can mean different things
- Tone, sarcasm, and culture matter

Course Map

From basic language processing to modern language AI systems

Unit I	**Language basics and NLP pipeline**	clean text, tokenize, POS, NER
Unit II	**Feature extraction**	BoW, TF-IDF, N-grams, Naive Bayes
Unit III	**Word embeddings**	Word2Vec, GloVe, semantic similarity
Unit IV	**RNNs and sequence models**	language models, LSTM, GRU
Unit V	**Machine translation and modern NLP**	Seq2Seq, BERT, GPT, conversational AI

Syllabus

Units I and II: from language basics to features

Unit I: Introduction to Language

Evolution of natural language

Introduction and need of NLP

Applications: discourse, dialog, opinion mining

NLP phases and pipeline

Preprocessing, tokenization, stemming, POS, normalization, NER

Unit II: Feature Extraction

One-hot encoding

Bag of Words, skip-grams, CountVectorizer

TF-IDF

Probabilistic language modeling

Naive Bayes, Markov models, N-grams, smoothing

Syllabus

Units III to V: from embeddings to modern language systems

Unit III: Word Embeddings

Word2Vec and GloVe

Gensim examples

Word senses and word windows

Neural networks basics

Dependency parsing; negative sampling

Unit IV: RNNs

Recurrent Neural Networks

Language models

N-gram connection

Vanishing gradients

LSTM and GRU

Unit V: Machine Translation

Seq2Seq machine translation

BERT, LaMDA, GPT

Conversational AI

Recent NLP papers

Responsible language AI

Unit I Real-World Examples

Introduction to Language and NLP pipeline

- YouTube captions: speech becomes text, punctuation is restored, names are recognized.
- WhatsApp autocorrect: tokenization, normalization, and context-based suggestions.
- Swiggy/Zomato review mining: opinion words reveal customer satisfaction.
- Bank KYC forms: named entity recognition finds names, dates, addresses, and IDs.
- Customer-support bots: intent detection decides whether a user wants refund, tracking, or complaint help.

Unit II Real-World Examples

Feature extraction: turning text into numbers

- Spam filtering: words such as 'winner', 'urgent', and suspicious links become useful features.
- News categorization: TF-IDF highlights words that separate politics, sports, finance, and technology.
- Search engines: query terms are matched with documents using weighted word statistics.
- Helpdesk routing: complaint text is converted into features and sent to the correct department.
- Sentiment dashboards: product reviews are converted into measurable signals.

Unit III Real-World Examples

Embeddings: meaning as vectors

- Semantic search: 'laptop battery issue' can match 'notebook power problem'.
- Resume-job matching: similar skills are recognized even when wording is different.
- Recommendation systems: similar product descriptions are grouped together.
- Duplicate question detection: two differently worded questions can have the same meaning.
- Chatbot retrieval: embeddings help find the closest answer from a knowledge base.

Unit IV Real-World Examples

RNNs and sequence modeling

- Keyboard prediction: the next word depends on previous words in the sentence.
- Speech recognition: audio is processed as a sequence over time.
- Chatbot response flow: earlier conversation context affects the next reply.
- Medical notes: symptoms mentioned across a paragraph must be connected.
- Older translation systems: LSTM and GRU models handled word order before transformers dominated.

Unit V Real-World Examples

Machine translation and modern NLP systems

- Google Translate and Microsoft Translator: cross-language communication at scale.
- Live captions and dubbing: speech, translation, and generation work together.
- Conversational AI: assistants answer, summarize, plan, and call tools.
- Document intelligence: legal, medical, and academic documents are summarized and queried.
- Research reading assistants: papers can be searched, summarized, compared, and explained.

Course Outcomes

What you should be able to do by the end

- **CO1** Explain evolution, need, applications, and phases of NLP.
- **CO2** Apply feature extraction methods such as BoW, TF-IDF, N-grams, and probabilistic models.
- **CO3** Use and compare word embeddings and neural representations.
- **CO4** Analyze RNN/LSTM/GRU and advanced NLP models for language tasks.
- **CO5** Understand machine translation and discuss current NLP research systems.

Evaluation Plan

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CA3 Case Studies

CO4 and CO5

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12 **marks**

Tentative: 22/10/2026

Build a small NLP Repo of Examples while learning. It makes CA3 easier.

Who Am I?

A short introduction before we begin

Dr. Sucheta Kolekar

- Teaching and research experience: 19+ years
- Research areas: AI/ML in HealthTech and EdTech, HCI, Cloud Computing
- Course faculty: Dr. Sucheta and Prof. Mayur
- Contact: sucheta.kolekar@sitpune.edu.in | 9008773043

Class style

- Examples first, theory next
- Ask why the model made a mistake
- Connect every unit to a real app or system

Top 10 NLP APIs

Google Cloud Natural Language API
Sentiment & Entity Analysis

Microsoft Azure Text Analytics
Key Phrases & Language Detection

OpenAI API
Text & Conversational AI

spaCy API
Linguistic Parsing

Amazon Comprehend
Entity Recognition & Sentiment

MeaningCloud API
Topic Extraction

IBM Watson NLU
Emotion & Keyword Analysis

AYLIEN Text Analysis
News & Content Classification

Hugging Face API
Transformers & Summarization

MonkeyLearn API
Custom Text Models

Platform	Best For	Free
Hugging Face	Pre-trained NLP models, inference, fine-tuning	✅
Google Colab	Running Python notebooks without installation	✅
Kaggle Notebooks	NLP datasets, competitions, notebooks	✅
spaCy	Industrial NLP pipeline	✅
NLTK	Learning NLP concepts from scratch	✅
OpenAI Playground	Prompt engineering and LLM experiments	Limited free
Cohere Playground	Text generation, embeddings	Limited free
IBM Watsonx.ai	Enterprise NLP and GenAI	Free tier
Google AI Studio	Gemini models and prompt engineering	Free tier
Azure AI Foundry	Enterprise NLP and LLM deployment	Free credits
Amazon SageMaker Studio Lab	ML and NLP experimentation	✅
Papers with Code	Latest NLP research with implementations	✅
TensorFlow Hub	NLP models and embeddings	✅
PyTorch Hub	Deep learning NLP models	✅
Gradio	Build NLP web applications	✅
Streamlit	Deploy NLP demos quickly	✅

NLP Pipeline

A simple mental model for the whole subject

Raw text → Cleaning → Tokens → Features / embeddings → Model → Output → Evaluation/Validation

Example

Review: "The phone battery is amazing."

Output: Positive sentiment

Same pipeline idea powers search, chatbots, translation, and summarization.

Modern systems add retrieval, tools, multimodal inputs, and safety checks.

Text	Representation	Model	Output	Evaluation
cleaning, tokens	features, embeddings	ML, RNN, transformer	class, answer, translation	accuracy, safety, trust

- Do not memorize only algorithms; ask what language problem each algorithm solves.
- Compare old methods and new methods: rules, statistics, embeddings, transformers, LLMs.
- For every tool, ask: What data? What representation? What mistakes? What risk?

Recent Advanced NLP

What has changed after the transformer and LLM era

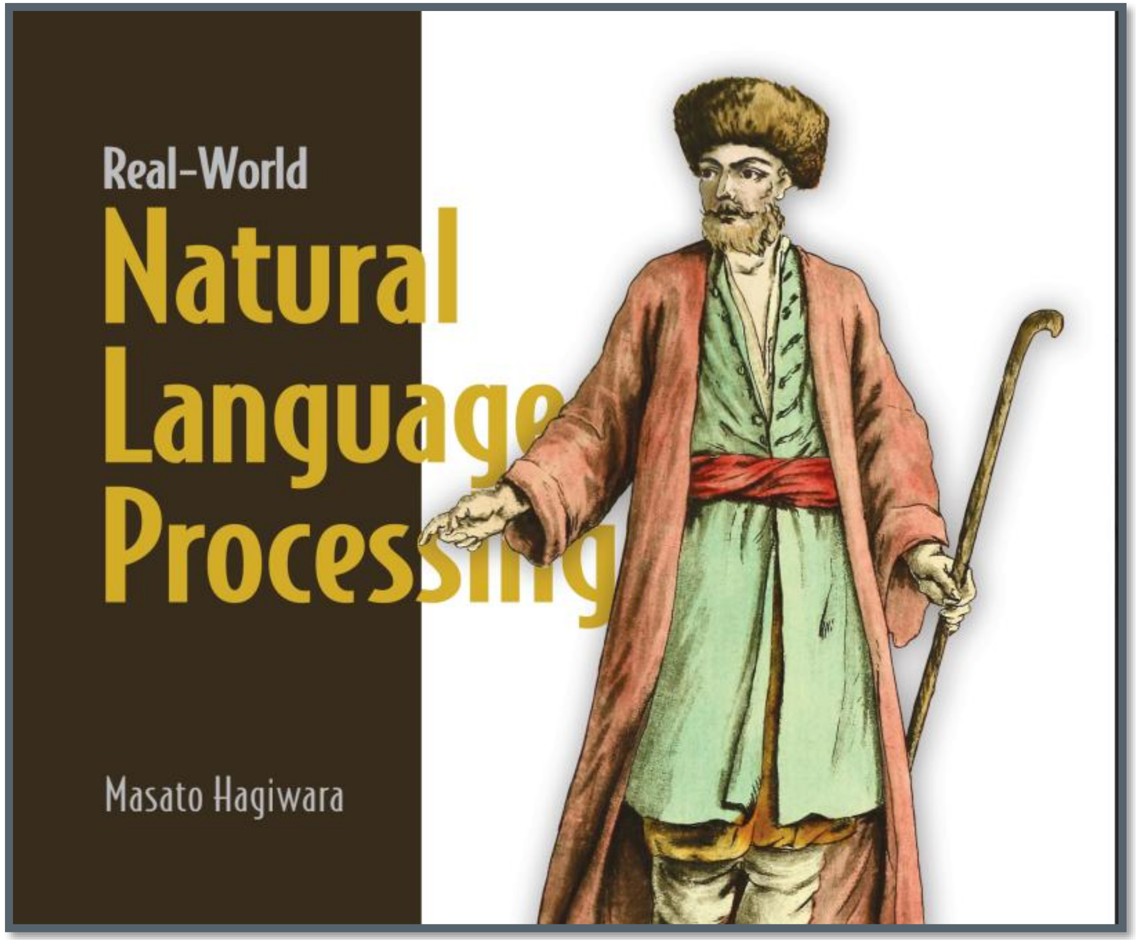
Transformers	Attention-based models became the foundation for BERT, GPT, T5, and modern LLMs.
Instruction LLMs	Models follow natural-language instructions for writing, coding, tutoring, and analysis.
RAG	Retrieval connects a model to documents before answering, useful for college, company, legal, and medical data.
Tool-using agents	Models can search, call APIs, run code, and complete multi-step tasks.
Multimodal AI	Text is combined with image, speech, video, PDFs, tables, and code.
Long-context models	Systems can read large documents, reports, repositories, or meeting transcripts at once.

More Advanced Topics to Watch

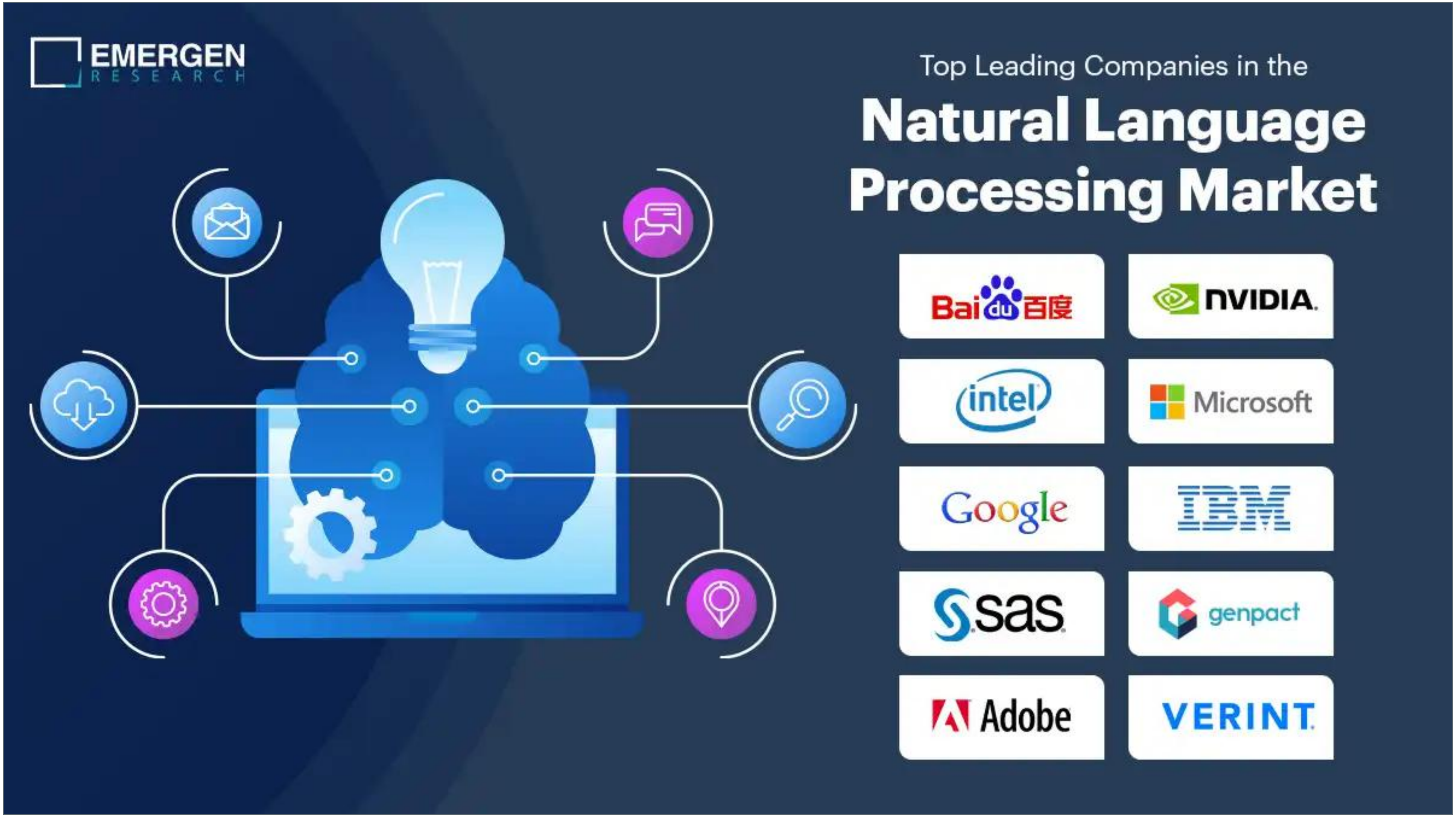
These are becoming important in industry and research

- Small language models: cheaper, faster, easier to deploy on private or edge systems.
- Domain-specific LLMs: models tuned for law, medicine, finance, education, and coding.
- Evaluation beyond accuracy: hallucination, faithfulness, bias, toxicity, latency, and cost.
- Responsible AI: privacy, copyright, prompt injection, misuse, and explainability.
- Indic NLP: multilingual and low-resource language support for Indian languages.
- Synthetic data: generated data is used to train and evaluate NLP systems, but quality control matters.

Reference Books



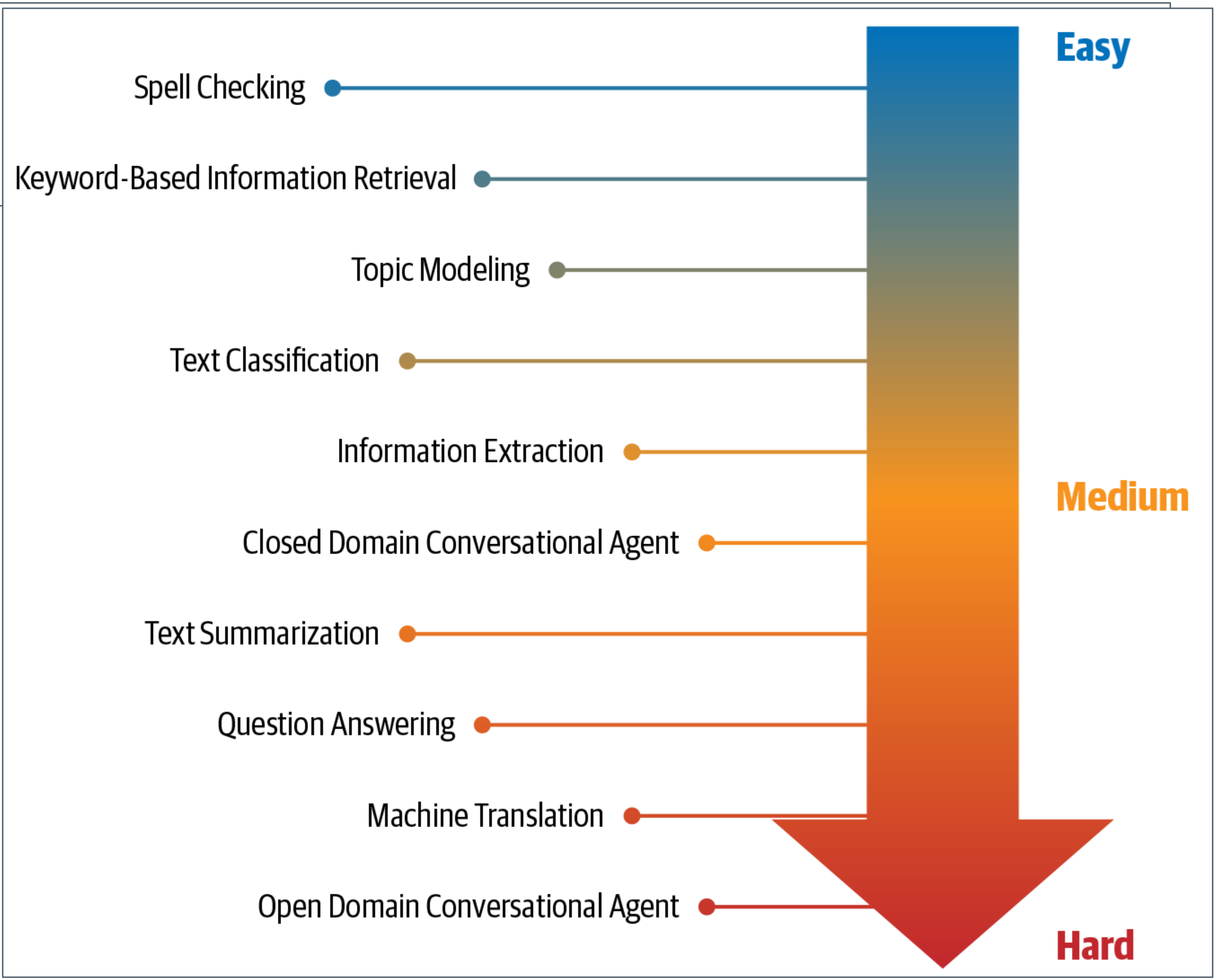
Rank	Research Lab	University
1	Stanford NLP Group	Stanford University
2	MIT Natural Language Processing Group	Massachusetts Institute of Technology
3	Berkeley AI Research (BAIR)	University of California, Berkeley
4	Allen Institute for AI (AI2)	Independent Research Institute
5	EdinburghNLP	The University of Edinburgh
6	UKP Lab	Technical University of Darmstadt
7	Princeton NLP	Princeton University
8	University of Washington NLP Group	University of Washington
9	EPFL NLP Lab	EPFL
10	Computational Linguistics and Information Processing (CLIP) Lab	University of Maryland, College Park


EMERGEN
RESEARCH
Top Leading Companies in the
Natural Language Processing Market
Baidu 百度
NVIDIA
intel
Microsoft
Google
IBM
sas
genpact
Adobe
VERINT

Top 10 NLP jobs in India (2026)

Rank	Job Role
1	LLM Engineer
2	Generative AI Engineer
3	NLP Engineer
4	AI Agent Developer (Agentic AI)
5	RAG Engineer
6	AI Research Engineer (NLP/LLMs)
7	Conversational AI Engineer
8	AI Solution Architect (GenAI)
9	AI Product Manager (GenAI)
10	Responsible AI / AI Safety Engineer

NLP Tasks



Easy
Spell Checking
Keyword-Based Information Retrieval
Topic Modeling
Text Classification
Information Extraction
Medium
Closed Domain Conversational Agent
Text Summarization
Question Answering
Machine Translation
Open Domain Conversational Agent
Hard

NLP ENGINEER: EVOLUTION FROM 2020–2023 TO 2026

From Traditional NLP to LLM-Powered Intelligent Systems

Traditional NLP (2020–2023)	VS	Modern NLP Engineer (2026)
Tokenization	→	LLM Applications
POS Tagging	→	RAG Systems
NER	→	AI Agents
Text Classification	→	Multimodal AI
Word2Vec	→	Foundation Models
LSTM	→	Transformers & LLMs
Machine Translation	→	Enterprise Copilots
Sentiment Analysis	→	Agentic Workflows
TF-IDF	→	Vector Embeddings
Rule-based Chatbots	→	Autonomous AI Assistants

KEY RESPONSIBILITIES OF MODERN NLP ENGINEER (2026)

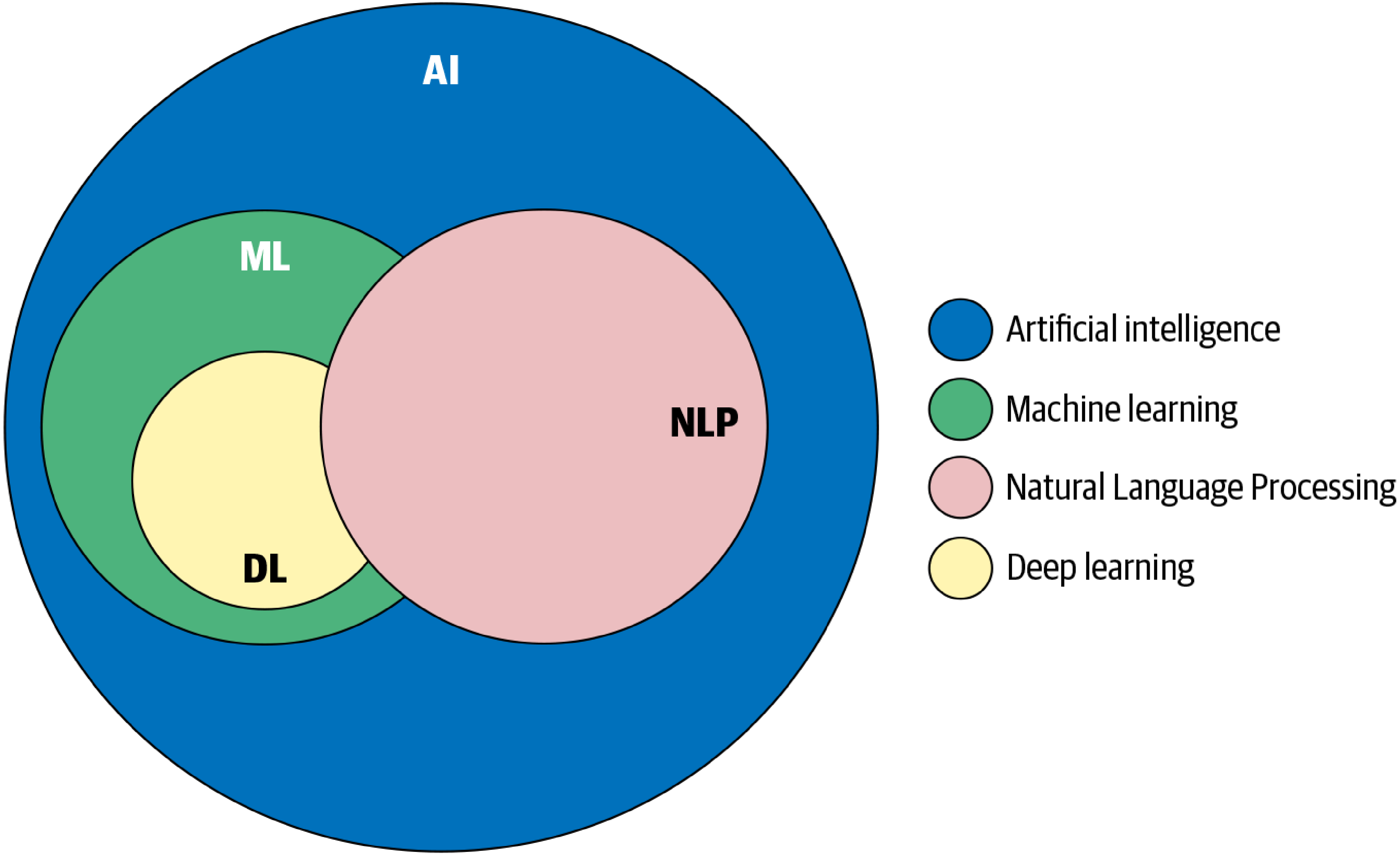
BUILD LLM-POWERED APPLICATIONS	Design, fine-tune and deploy Large Language Models for real-world products and user experiences.
DESIGN RAG PIPELINES	Build retrieval-augmented generation systems using vector databases and enterprise data.
DEVELOP AI AGENTS	Create intelligent agents that can reason, plan, use tools and automate complex tasks.
BUILD MULTIMODAL AI SOLUTIONS	Work with text, image, audio, video and structured data to build richer AI experiences.
FINE-TUNE & OPTIMIZE FOUNDATION MODELS	Use advanced tuning techniques (LoRA, QLoRA, PEFT) to adapt models efficiently for specific domains.
WORK WITH VECTOR DATABASES	Implement and optimize vector search using FAISS, Pinecone, Milvus, ChromaDB and other tools.
EVALUATE & IMPROVE MODEL QUALITY	Build robust evaluation frameworks for accuracy, factuality, bias, toxicity, hallucination and safety.
DEPLOY SCALABLE AI SOLUTIONS	Deploy models as scalable APIs and microservices using Docker, Kubernetes and CI/CD pipelines.
ENSURE RESPONSIBLE & SAFE AI	Implement guardrails, privacy measures, fairness checks and follow responsible AI best practices.
COLLABORATE & DRIVE BUSINESS IMPACT	Work with cross-functional teams to integrate AI solutions and deliver measurable business value.

CORE SKILLS IN 2026

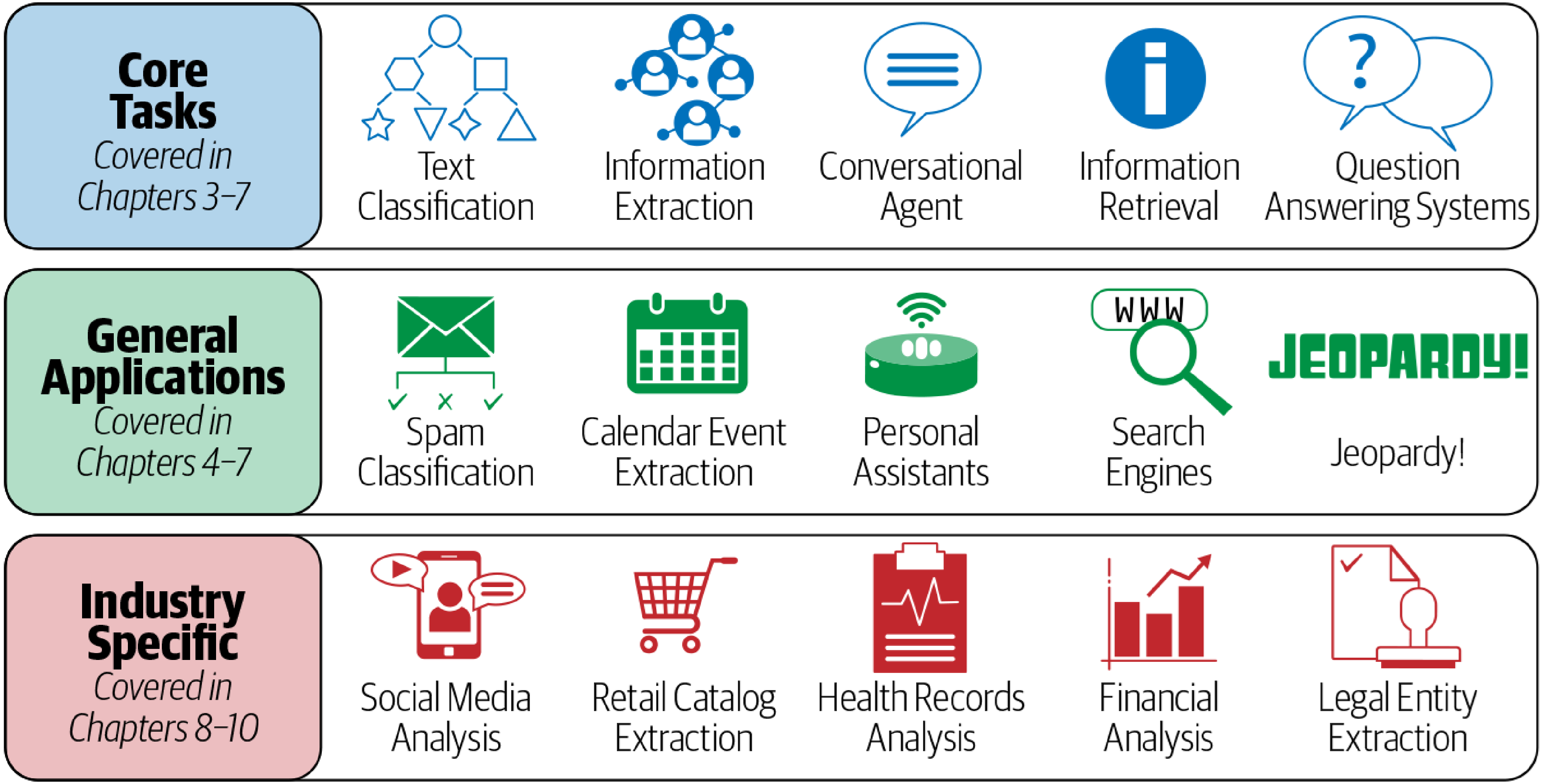
- Python
- Hugging Face Transformers
- PyTorch
- LLMs (OpenAI, Llama, Mistral, Gemma, Claude, etc.)
- LangChain LlamaIndex LangGraph
- Vector Databases (FAISS, Pinecone, Milvus, ChromaDB)
- AI Agents (CrewAI, AutoGen, Semantic Kernel)
- Cloud & MLOps (AWS, Azure, GCP, Docker, K8s, MLflow)
- Prompt Engineering & Model Evaluation
- Responsible AI & AI Safety

2026 NLP ENGINEER = LLM EXPERT + AI BUILDER + PROBLEM SOLVER + IMPACT CREATOR

Machine Learning, Deep Learning, and NLP



Applications of NLP



Modern NLP System Example

College chatbot over official course documents

Student question

"When is CA1 and what unit is included?"

What happens inside

Retrieve relevant course file pages

Check dates and unit mapping

Generate answer with source context

Warn if source information is missing

Show source for verification

This is NLP + search + reasoning + evaluation, not just a chatbot.

Mini Activity

Human NLP pipeline in five minutes

Sentence: "The food was late, but the support team solved it quickly."

1. Find important words
2. Detect sentiment words
3. Notice contrast after 'but'
4. Decide final sentiment
5. Explain your reasoning

Takeaway: NLP is hard because meaning is not just a bag of words.

Brainstorming Activity 1

Warm-up MCQs: spot the NLP around you

1. Which feature is the clearest example of NLP?

A. Increasing phone brightness

B. Translating Hindi text to English

C. Compressing an image

D. Sorting files by date

Answer: B

Brainstorming Activity 1

Warm-up MCQs: language understanding vs pattern matching

2. A chatbot answers from a company's policy PDF. What modern NLP pattern is most likely being used?

- **A. RAG: retrieve relevant passages, then generate an answer**
- **B. Only image compression**
- **C. Manual copy-paste by a human**
- **D. Random sentence generation**

Answer: A

Brainstorming Activity 1

Warm-up MCQs: where embeddings help

3. Why can a search engine understand that 'car' and 'automobile' are related?

- **A. It stores words only alphabetically**
- **B. It learns vector representations of meaning**
- **C. It ignores synonyms**
- **D. It reads only the first letter**

Answer: B

Mini Activity 2

Human NLP pipeline in five minutes

Task: classify the sentence below as Positive, Negative, or Neutral.

"The movie started slowly, but the ending was brilliant."

Tokenize the sentence.

Remove noise if needed.

Find important words.

Handle contrast after the word 'but'.

Decide the final sentiment.

why is this harder than just counting positive and negative words?